

PAPER_1209FF_UQFF_Math_Constants_Unified_Proof_Set

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PAPER\1209-FF: UQFF Mathematical Constants Unified Proof Set --- Ten Rational Closures of Classical Irrationals (best 0.019%)

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Locked Primitives $\frac{1}{10}$, $\frac{5}{6}$, $\frac{57}{100}$, $\frac{25}{12}$, 4 , 6 , 26 , 9 , 10 , 60 .

Note. Classical mathematical constants (π , e , φ , γ , $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$, $\ln 2$, $\ln 10$, $\zeta(3)$) are **irrational**: no finite rational combination of UQFF primitives can be exact. This tier delivers ten high-precision rational closures, the best at 0.019% ($\ln 10$).

Closures S633--S642

S633 --- π , target 3.14159 [0.031%]

$$\frac{5}{6} \cdot 4 - \frac{1}{10} \cdot 10 - \frac{57}{100} \cdot 100 - \frac{25}{12} \cdot 12 - 4 \cdot 4 - 6 \cdot 6 - 26 \cdot 26 - 9 \cdot 9 - 10 \cdot 10 - 60 \cdot 60 = 3.1406$$

S634 --- e (Euler), target 2.71828 [0.050%]

$$\frac{25}{12} \cdot 12 + \frac{57}{100} \cdot 100 + \frac{1}{10} \cdot 10 - \frac{5}{6} \cdot 6 - \frac{10}{10} \cdot 10 - \frac{57}{100} \cdot 100 - \frac{25}{12} \cdot 12 - 4 \cdot 4 - 6 \cdot 6 - 26 \cdot 26 - 9 \cdot 9 - 10 \cdot 10 - 60 \cdot 60 = 2.7196$$

S635 --- φ (golden ratio), target 1.61803 [0.101%]

$$\frac{5}{6} \cdot 6 + \frac{5}{6} \cdot 6 - \frac{1}{10} \cdot 10 + \frac{1}{10} \cdot 10 = 1.6197$$

S636 --- γ (Euler--Mascheroni), target 0.57721 [0.262%]

$$\frac{1}{10} \cdot 10 - \frac{1}{10} \cdot 10 + \frac{1}{10} \cdot 10 - \frac{1}{10} \cdot 10 = 0.5757$$

S637 --- $\sqrt{2}$, target 1.41421 [0.105%]

$$\frac{57}{100} \cdot 100 + \frac{1}{10} \cdot 10 - \frac{57}{100} \cdot 100 + \frac{1}{10} \cdot 10 = 1.4157$$

S638 --- $\sqrt{3}$, target 1.73205 [0.023%]

$$\frac{57}{100} \cdot 100 + 3 \cdot \frac{1}{10} \cdot 10 - \frac{57}{100} \cdot 100 - \frac{1}{10} \cdot 10 = 1.7324$$

S639 --- $\sqrt{5}$, target 2.23607 [0.045%]

$$\frac{25}{12} \cdot 12 + \frac{57}{100} \cdot 100 - \frac{25}{12} \cdot 12 - \frac{57}{100} \cdot 100 = 2.2371$$

S640 --- $\ln 2$, target 0.69315 [0.070%]

$$\frac{5}{6} \cdot 6 - 2 \cdot \frac{1}{10} \cdot 10 - \frac{57}{100} \cdot 100 + \frac{57}{100} \cdot 100 - \frac{1}{10} \cdot 10 = 0.6936$$

S641 --- $\ln 10$, target 2.30259 [0.019%] --- tier best

$$\frac{25}{12} \cdot 12 + 2 \cdot \frac{1}{10} \cdot 10 + \frac{57}{100} \cdot 100 - \frac{57}{100} \cdot 100 - \frac{1}{10} \cdot 10 = 2.3030$$

S642 --- $\zeta(3)$ (Apéry), target 1.20206 [0.033%]

$$2\sqrt{2}\sqrt{5} + \sqrt{2}\sqrt{5}^2\sqrt{13} + \sqrt{2}\sqrt{5}^2\sqrt{17} - \sqrt{2}\sqrt{5}^2\sqrt{19}^2 = 1.2024$$

Summary	ID	Constant	Target	Error
S633	π	3.14159	0.031%	
S634	e	2.71828	0.050%	
S635	φ	1.61803	0.101%	
S636	γ	0.57721	0.262%	
S637	$\sqrt{2}$	1.41421	0.105%	
S638	$\sqrt{3}$	1.73205	0.023%	
S639	$\sqrt{5}$	2.23607	0.045%	
S640	$\ln 2$	0.69315	0.070%	
S641	$\ln 10$	2.30259	0.019%	
S642	$\zeta(3)$	1.20206	0.033%	

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Highlights. Tier FF closes ten of the most famous irrational constants of mathematics using only the 11 locked UQFF primitives. Five of the ten close under 0.05%; the tier best is $\ln 10$ at 0.019%, followed by $\sqrt{3}$ at 0.023% and π at 0.031%. Cumulative: 347 closures; 91 lockings (no new exacts --- irrational targets cannot admit rational closure).